can just sit back and relax on your sofa, while you organise some virtual paintings on your wall, do some shopping online or play game. and if you think sitting back and moving your hands is going to make you lazy, think again. It gets lazier! The next step is interaction through facial movements. Software Developer Eye Tribe is about to release an eye-tracking software for Android mobiles where users can navigate their mobile, play games such as Fruit Ninja with just eye-movements!

Interaction with facial recognition has a major application in the video games industry, and some developers have already implemented this technology to improve their game's mechanics. The most widely played MMORPG – World of Warcraft (WOW), which already immerses players in their beautiful fantasy worlds are taking the experience a step forward. With the help of a facial recognition and animation tool called "FaceShift", Blizzard (creator of WOW) will capture your facial expression and animate your character accordingly.

Voice-based interfaces

Voice Recognition has always been in the pipeline since ages;but always faced problems due to varied accents and a lot of environmental issues. But due to the advanced parsing algorithms and computational power, voice-based interfaces are a reality now. One of the best recent examples is "Siri" for iPhone. It accepts Voice commands, suggests you places to visit, books your calendars, handles your phone on your command and also can converse with you. Before Siri, Google also released an app for the iPhone accepted voice commands and could be used to do Google searches. In May 2013, Google released a conversational voice search function along with the new version of Chrome browser. So, now you can talk to the search engine directly; ask questions like "Who's the 25th president of USA?" It also introduced semantic search, because of which needless information can be avoided. Google's answer to the question – "When was he born?" would be the birthdate of William McKinley, who was the 25th president of the US. Google Glass, the head-mounted wearable computer also uses Voice commands as input. Users can now basically talk to their glasses to take pictures, upload to Facebook or send an email on the fly.

Airborne displays

Remember how Tony Stark catches a digital file, crushes it and throws it in the garbage bin in the movie "Iron Man". Remember how he holds the Stark Expo model map and zooms and turns the model with his bare hands? That's an example of airborne displays, and we should have them soon enough. A Finnish company FogScreen presented it's mind-blowing technology in CES – a display system that projects images onto a curtain of fog. So apps and games run on a layer of vapour and you can use your hands to interact with the digital content.

Displair, a Russia-based technology firm has added 3D multi-touch capabilities making images and models manipulatable in three dimensions in open space. As the digital content is in the air, it is penetrable.

A user operating a 3D model in Displair

Brain-to-machine

As if gesture-based, voice-based and airborne Interfaces are not enough, scientists are developing what might be called the ultimate computer interface – interaction of your brain directly with a machine. It doesn't get simpler than this; imagine turning your lights off just by thinking about it, or instructing your personal robot to bring a lemonade just by thinking! This is in line with the concept of the movie Avatar – where a human handles a humanoid with his/her mind. This has numerous applications. If this technology is combined to robotic arms or humanoids, it can be used to help people with paralysis and other disabilities in their daily activities. Researchers at Brown University developed a robotic arm which could respond to the brain activity of a paralyzed woman. The robotic arm when commanded, could pour her coffee for her, and then return to rest on the table.

Virtual Fitting Room : Women trying dress using Virtual Fitting Room

Organisations such as Samsung are working on technologies that can help users operate their phone with their minds. The idea is that in the near future, you will not need to take out your phone from your pocket to answer a call, dial a contact, read a message or even to tell your phone what song you want to hear. The US government is investing heavily in the BRAIN initiative which focuses on research projects through innovative neuro technologies. One of the current projects is to create a brain-machine interface with the help of which soldiers can control a jet or other weapon systems like in the 1982 film "FireFox".

Every new technology has a major application in video games, particularly this one. Imagine playing a game where you dodge bullets or play temple run just with your mind. NeuroSky, a california-based startup is onto this and has made decent progress.

This technology can also be used to train the brains of mentally-challenged people. But, the brain-to machine interface brings in a lot of privacy concerns. If the machine can understand people's thought process, then it might be too risky as that information can also be hacked. Another concern is regarding how contextually aware the interface is. For example, if you were just thinking about *biryani* but are not really hungry, how would your personal robot